



DECISION 618: Data Analytics for Business

Fuqua School of Business, Duke University

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Course Overview

This course will investigate how data can be used to directly impact business decisions. The volume, variety and velocity associated with currently available data motivate an important paradigm for data-driven decision-making in business. Beyond extracting information from data, and creating data-driven real-time autonomous decision making systems, the data itself should be thought of as an asset for current and future use.

In this course, we will study the core ideas behind data science, challenges associated with data, and the interplay between data science and business decisions. We will focus on (i) general principles that are long lasting despite of the rapid changing technology; (ii) specific algorithms/technologies that are currently being used in many industries; and (iii) “hands-on” analyses of actual datasets to develop methodologies. During classes we will discuss a variety of real-life examples such as customer retention, health risk prediction, social media analysis, online advertisement, and renewable energy operations.

Ultimately, the course aims to develop “data-analytic” thinking. This should enable you to evaluate how data can improve performance, identify opportunities, and assist in decision making for managers. The understanding of the principles of data science techniques will allow you to interact effectively with experts/consultants, and to quickly conduct pilot studies.

Prerequisites and Audience

The course is designed for students with a solid understanding of probability and statistical concepts. All Fuqua students that have passed the core statistics requirement DECISION 610: Foundations of Business Analytics (or were exempt/conditionally exempt) have the necessary background for this course. (Exchange students should contact the instructor prior registering to the course.) The insights developed throughout the course should benefit everyone in a management role.

Topics and Schedule

SESSION	TOPIC	Data/Application
1	Introduction, Google's Search Engine, and network Analytics	
2	Linear Regression, training/testing split	Bluebike, Solar Power
3	Logistic Regression for classification	Customer Churn
4	CART for Regression and Classification, cross validation	Medical Costs Parole decisions
5	Random Forests and Boosting	Click-Thru-Rates
6	Feature Engineering and regularization	NYC Taxi, Diabetes detection
7	Clustering and Collaborative Filtering	Customer Segmentation Recommender Systems
8	Neural Networks	Image and text recognition
9	Guest speaker: Phil Keenan on Wednesday 9/30 at 8am <i>Vehicle Content Optimization (VCO) at General Motors</i>	
10	Optimization for Contract Design	Drug development Auction design
11	Guest speaker: Alexei Robsky from Microsoft <i>Data Science and AI at Google and Microsoft</i>	
12	Student Project Presentations	

Computing and Software

The course will have a “hands-on” component. The available data sets and GenAI-supported coding environment, entry barrier for computing is much lowered compared with a mere few years ago. We will use state-of-the-art implementations, mostly in Python, and some in R.

The course is designed to meet the needs of those with no prior programming experience. We will focus on the fundamental ideas behind those algorithm, and insights generated from running code, not coding itself.

Recent developments in GenAI makes many computing tasks much easier to complete. Efficient use of GenAI tools relies on understanding of the concepts. We will heavily utilize the GenAI tools in coming up with the computer code.

After each class you will receive the Python/R code that delivers most (if not all) of what we cover in class. This code will allow you to get started in analyzing the data in homework and your project. Running and adapting these codes is an important learning component.

In addition to the documentation and files provided online, we will also hold TA office sessions that will help with coding as well. (Details will be announced.)

Boot camp on Python (and a little R) in Google Colab

We want to focus on data science concepts and its use. Even though the course includes work involving coding in computer languages Python and R (in team assignments and the final project), the course is not about and does not teach you how to create and debug programs. I encourage students to participate the “Boothcamp on Python in Google Colab” offered by Prof. Salman Azhar to learn the foundations of generating and debugging code in Python and R in the Google Colab environment. This is not a class on coding, but you need to understand the basics to take advantage of the tools and know when AI hallucinated!

Course Grading

Letter grades will be assigned in accordance with Fuqua's recommended grade distribution for elective courses. The relative contributions of different aspects of the course are as follows:

- final project (team) 35%
- homework assignments (individual) 50%
- class participation (individual) 15%.

There is an individual homework assignment each week. For these homework assignments you are allowed to work within your team of 3 and use GenAI tools to help with coding. However you are not allowed to directly share files. Anything you submit for these homework assignments needs to be generated by yourself (with help from AI and discussions within your team). Because I allow and encourage you to use AI tools for the homework assignment, grading of homework is only based on proper completion.

Late submissions of the weekly homework will be penalized 20% per day late. If you anticipate that you may be unable to meet a deadline, discuss it with your instructor in advance; extensions require advance approval from your instructor.

The final project asks you to develop your own data science application. Planning should begin as soon as possible. Projects can range through a variety of practical applications. See Canvas for more details. The final project is a team project. Your team's work must reflect the collective effort of the designated team. Discussion or collaboration with people outside the team, copying or adapting other teams' work (including from prior years), or sharing your team's work outside the team without your instructor's explicit authorization are not permitted. All team members share responsibility for the integrity of the submission. All team members in turn receive the same grade.

Class Participation and Conduct

Attendance is required in *every* class session. Starting from the second missed class, each missed class carries a 5% reduction in the grade, regardless of reasons. Missing 4 or more class sessions results in automatic course failure.

Check in using the Fuqua Check-in app at the start of each session and report any early departure from class via the app. Misreporting attendance is an Honor Code violation.

Your class participation will be evaluated subjectively, but will rely upon measures of punctuality, attendance, and the relevance/insight of class participation. Your class participation will be judged by what you add to the class environment, regardless of your technical background. In general, questions and comments are welcome and encouraged.

Consistent with Fuqua Classroom Norms, we have the following policies:

- Students are expected to arrive promptly and be ready for class to start on time and to stay for the entire class.
- Laptops, e-pads, and smartphones are not to be open in the classroom, except for note-taking on a flat screen.
- Maintenance of a professional atmosphere by using respectful comments and respectful humor.
- Refraining from distracting or disrespectful activities (e.g., avoiding side conversations and games).
- Courtesy towards all participants in the classroom.
- Observance of the most conservative standards when one is unsure about which norms apply.

Please refer to the Fuqua Classroom Norms materials for more details (see below). Violations of these policies will be marked. Three or more violations will result in an automatic penalty of a letter grade. We ask that you use a name card for all class sessions.

Honor Code

The Fuqua School of Business Honor Code applies to all aspects of this course. Violation of the Honor Code is a serious offense, and suitable consequences include grade reduction, an F grade, a transcript notation, delay of graduation, or expulsion.

The objective here is to learn. In our experience, the material of this class is best learned through individual practice and exposure to a variety of application contexts.

Classroom Etiquette and Fuqua Classroom Norms

Keep in mind that the Fuqua Classroom Norms apply. In particular, these norms imply: (1) Class will start on time and end on time; breaks will end at the time specified. Students should be in their seats and prepared for class discussion before class starts and before the break ends. (2) Some classes may have handouts. You should arrive in time to pick up a handout at the front of the class or turn in an assignment. Do not walk to the front once class is started. (3) Late arrivals should take a seat at the rear of the classroom. This will avoid disruption to others. It is fine to move into a better seat at the break. (4) Do not leave the classroom while class is in session. Wait until a break or until class ends. If you are ill, or awaiting news because of a family emergency, let the instructor know before class and sit at the back of the classroom. (5) No electronic devices are permitted in class. In the event you need to be in touch because of a family emergency, let the instructor know before class and keep your electronic device on vibrate. The complete description of the Fuqua Classroom Norms can be found at the course site.